

Question number	Answer	Mark
1(a)(i)	B human immunodeficiency virus (HIV) and tobacco mosaic virus (TMV)	(1)

Question number	Answer	Mark
1(a)(ii)	D tobacco mosaic virus (TMV) only	(1)

Question number	Answer	Mark
1(a)(iii)	D protein	(1)

Question number	Answer	Additional guidance	Mark
1(a)(iv)	• calculation of volume of TMV (1) • calculation of size difference (1) Example of calculation: $3.14 \times 40 \times 40 \times 300 = 1\,507\,200$ 19 / 19.4 / 19.42	Allow ecf Correct answer with no working shown gains full marks	(2)

Question number	Answer	Mark
1(b)(i)	• reverse transcriptase and integrase	(1)

Question number	Answer	Mark
1(b)(ii)	An explanation that includes the following points: • because these enzymes are not present in the host cell (1) • because reverse transcriptase is needed to make a DNA copy of the viral RNA (1) • because integrase is needed to incorporate the DNA copy into the host cell genome (1)	(3)

